

QC HISTORY

Historical QC performance

Report number

HIS-20261005-33D28471

Issue date

5 October 2026

Report scope

Historical quality control

Issued by

Laboratório de Paleoceanografia e Paleoclimatologia

Demonstration data and simulated approvals

Individual QC observations are grouped by method version, material lot and intervention period. Descriptive SD is not an approved intermediate-precision estimate until the period is reviewed.

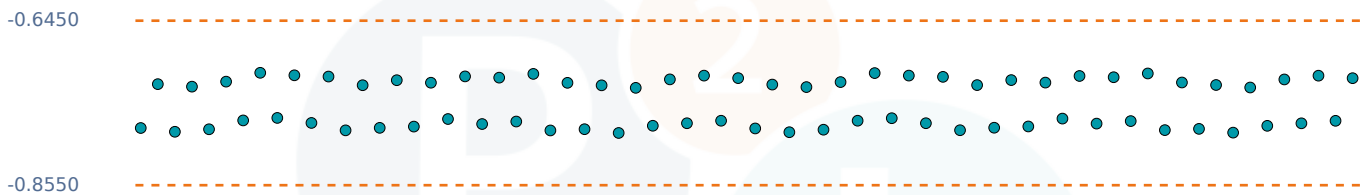
SHP2L · method v1 · 12 runs

Lot: DEMO-2026-01

DEMO Kiel IV / MAT253 Plus / Dual Inlet, configuration A · Period: initial

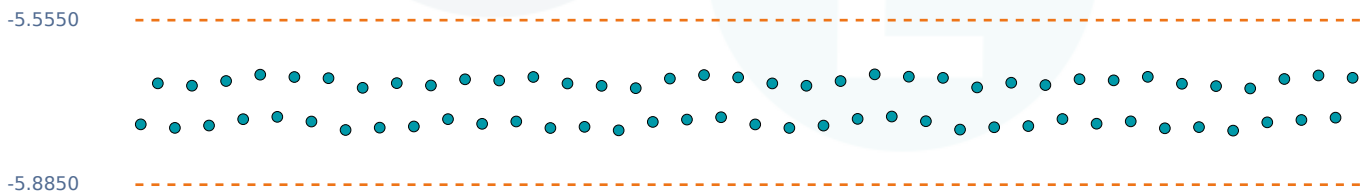
Isotope	Observations	Mean / ‰	SD / ‰	Control status
$\delta^{13}\text{C}$	72	-0.7499	0.0298	no signal detected
$\delta^{18}\text{O}$	72	-5.7201	0.0438	no signal detected

$\delta^{13}\text{C}$ / ‰



Individual QC observations, acquisition order

$\delta^{18}\text{O}$ / ‰



Individual QC observations, acquisition order

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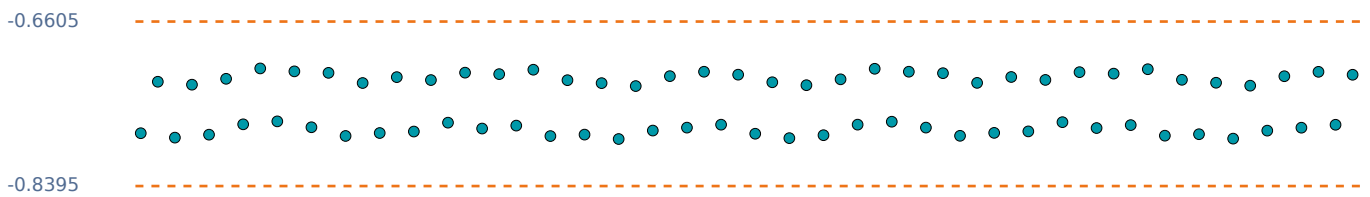
SHP2L · method v2 · 12 runs

Lot: DEMO-2026-01

DEMO Kiel IV / MAT253 Plus / Dual Inlet, configuration A · Period: initial

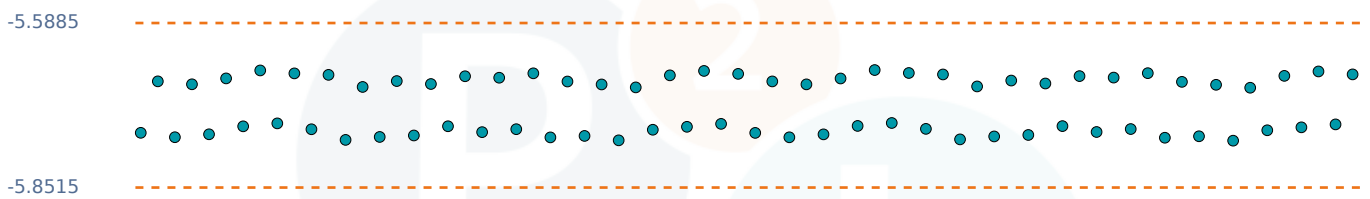
Isotope	Observations	Mean / ‰	SD / ‰	Control status
$\delta^{13}\text{C}$	72	-0.7499	0.0298	no signal detected
$\delta^{18}\text{O}$	72	-5.7201	0.0438	no signal detected

$\delta^{13}\text{C}$ / ‰



Individual QC observations, acquisition order

$\delta^{18}\text{O}$ / ‰



Individual QC observations, acquisition order

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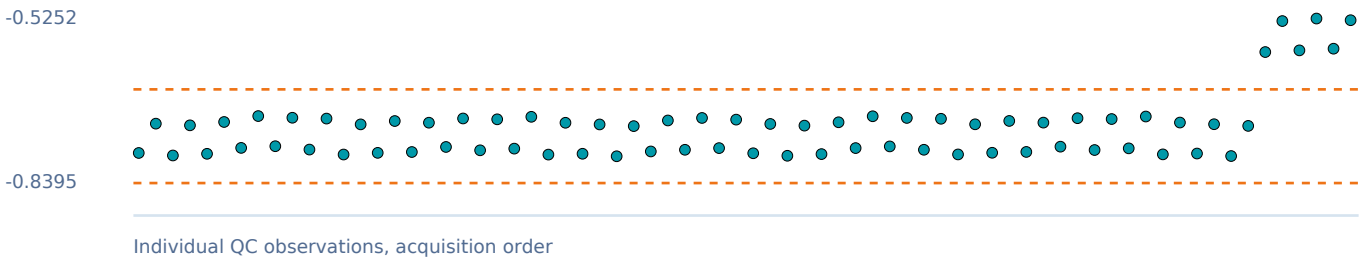
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SHP2L · method v3 · 12 runs

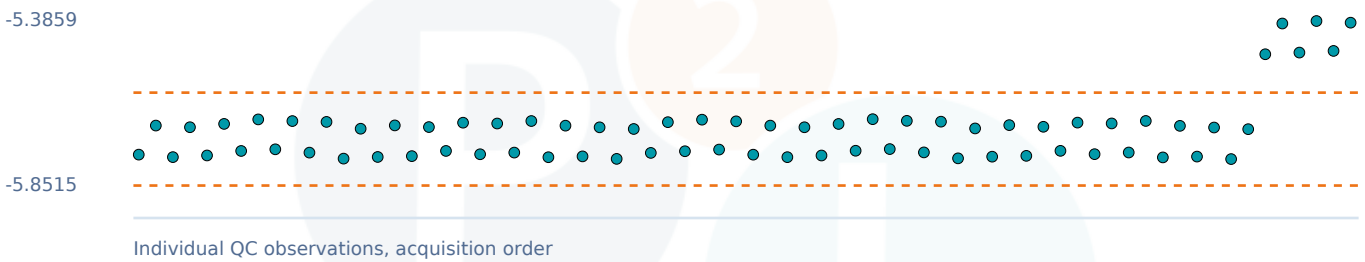
Lot: DEMO-2026-01
 DEMO Kiel IV / MAT253 Plus / Dual Inlet, configuration A · Period: initial

Isotope	Observations	Mean / ‰	SD / ‰	Control status
$\delta^{13}\text{C}$	72	-0.7341	0.0615	out of control
$\delta^{18}\text{O}$	72	-5.6968	0.0914	out of control

$\delta^{13}\text{C}$ / ‰



$\delta^{18}\text{O}$ / ‰



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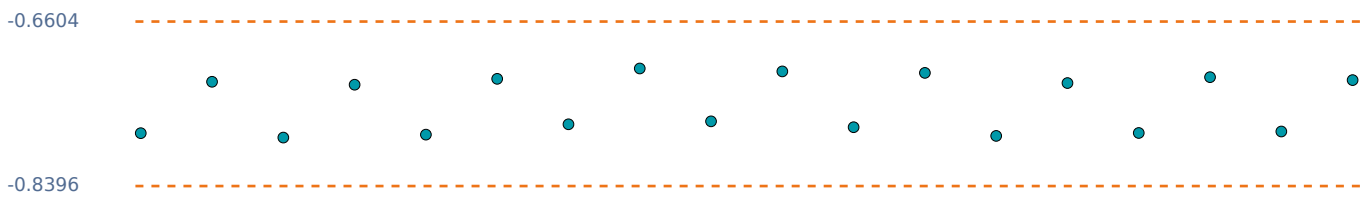
SHP2L · method v4 · 3 runs

Lot: DEMO-2026-01

DEMO Kiel IV / MAT253 Plus / Dual Inlet, configuration A · Period: initial

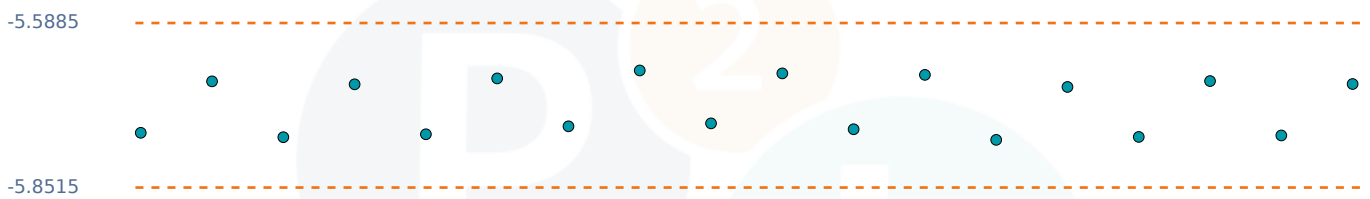
Isotope	Observations	Mean / ‰	SD / ‰	Control status
$\delta^{13}\text{C}$	18	-0.7505	0.0305	no signal detected
$\delta^{18}\text{O}$	18	-5.7214	0.0448	no signal detected

$\delta^{13}\text{C}$ / ‰



Individual QC observations, acquisition order

$\delta^{18}\text{O}$ / ‰



Individual QC observations, acquisition order

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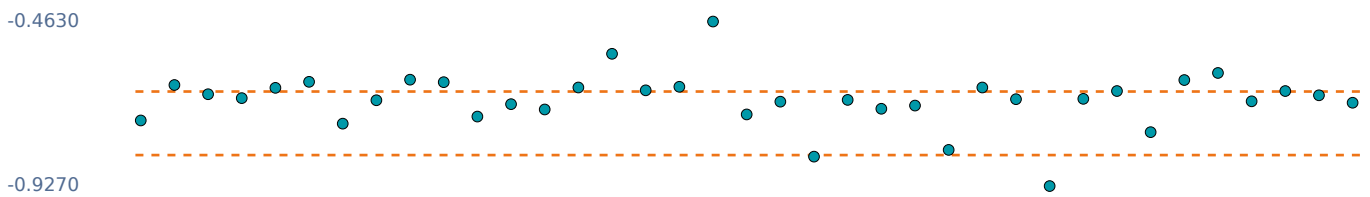
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SHP2L · method v4 · 1 runs

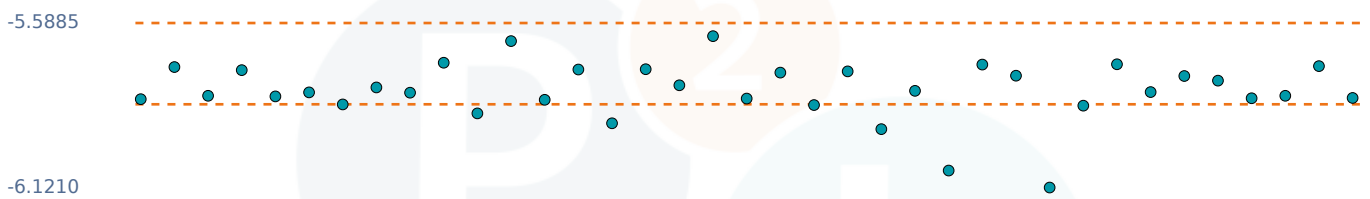
Lot: DEMO-2026-01
 DEMO Kiel IV / MAT253 Plus / Dual Inlet, configuration A · Period: initial

Isotope	Observations	Mean / ‰	SD / ‰	Control status
$\delta^{13}\text{C}$	37	-0.6842	0.0779	out of control
$\delta^{18}\text{O}$	37	-5.8069	0.0953	out of control

$\delta^{13}\text{C}$ / ‰



$\delta^{18}\text{O}$ / ‰



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SHP2L · method v4 · 33 runs

Lot: DEMO-2026-01
 DEMO Kiel IV / MAT253 Plus / Dual Inlet, configuration A · Period: initial

Isotope	Observations	Mean / ‰	SD / ‰	Control status
δ ¹³ C	217	-1.3300	2.9215	out of control
δ ¹⁸ O	217	-6.4013	2.1695	out of control

δ¹³C / ‰



δ¹⁸O / ‰



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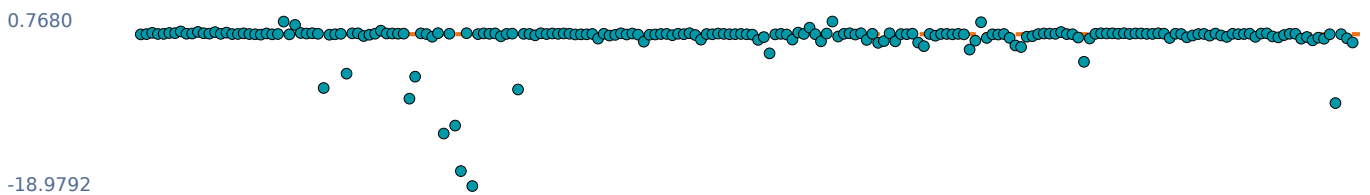
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SHP2L · method v4 · 19 runs

Lot: DEMO-2026-01
 DEMO Kiel IV / MAT253 Plus / Dual Inlet, configuration A · Period: initial

Isotope	Observations	Mean / ‰	SD / ‰	Control status
$\delta^{13}\text{C}$	213	-1.2902	2.2920	out of control
$\delta^{18}\text{O}$	213	-6.4164	2.9014	out of control

$\delta^{13}\text{C}$ / ‰



$\delta^{18}\text{O}$ / ‰



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Reviewed homogeneous periods

Period	Reviewer	Estimator and evidence
DEMO controlled period 1 (2026-04-05)	Demo scientist	SD of individual observations, not SE of mean Synthetic workflow demonstration; example evidence reviewed, not a laboratory authorization.
DEMO controlled period 2 (2026-06-04)	Demo scientist	SD of individual observations, not SE of mean Synthetic workflow demonstration; example evidence reviewed, not a laboratory authorization.
DEMO controlled period 3 (2026-08-03)	Demo scientist	SD of individual observations, not SE of mean Synthetic workflow demonstration; example evidence reviewed, not a laboratory authorization.
DEMO controlled period 4 (2026-10-02)	Demo scientist	SD of individual observations, not SE of mean Synthetic workflow demonstration; example evidence reviewed, not a laboratory authorization.

